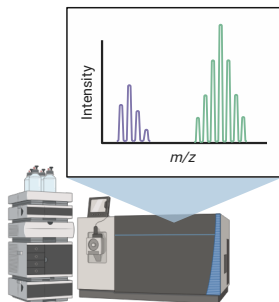
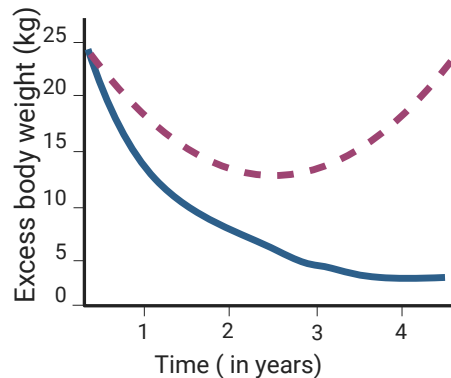


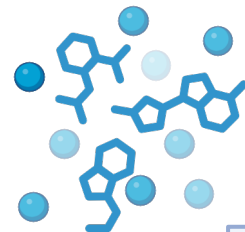
Discovery cohort,
LABS-2

Latent class modeling



Lipids
Amino acids
Carbohydrates
Xenobiotics
Nucleotide
derivatives
Other
metabolites

STUDY DESIGN

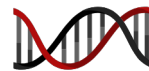


Metabolite
risk score
(MetRS)

Replication cohorts
Post -RYGB cohort
(n=36, median 7.2
years)

Obesity extremes
(n=198)

Genetic analyses



Framingham
Heart Study